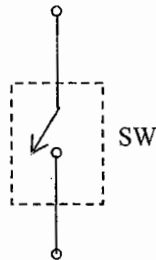


Definition:

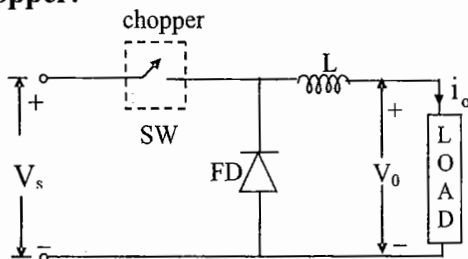
- A chopper is a static circuit that converts fixed dc input voltage to a variable dc output voltage directly. As choppers involve one stage conversion, these are more efficient.
- Chopper system offers smooth control, high efficiency, fast response and regeneration.
- The power semiconductor devices used for a chopper circuit can be
 - Force commutated thyristor.
 - Power BJT.
 - Power MOSFET
 - GTO (or)
 - IGBT.

These devices in general can be represented by a switch SW with an arrow.



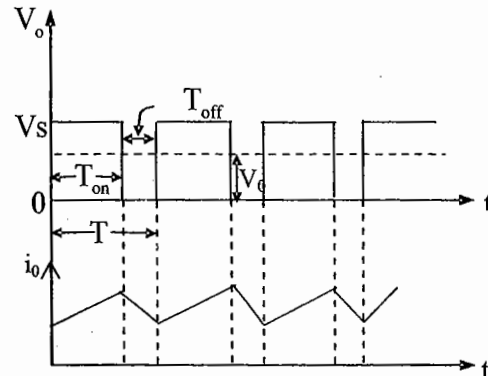
- Chopper are of two types
 - Step down chopper (or) Buck converter ($V_o < V_s$)
 - Step up chopper (or) Boost converter ($V_o > V_s$)

Principle of operation of STEP DOWN chopper:



- During the period T_{on} , chopper is on and load voltage is equal to source voltage V_s . During the interval T_{off} , chopper is off, load current flows through the free wheeling diode FD. As a result, load terminals are short circuited by FD and load voltage is therefore zero during T_{off} .

- During T_{on} , load current rises where as during T_{off} Load current decays



- Average load voltage V_o is given by

$$V_o = \frac{T_{on}}{T_{on} + T_{off}} V_s = \frac{T_{on}}{T} V_s = \alpha V_s$$

$$V_o = \alpha V_s$$

Where, T_{on} = on time; T_{off} = off time.

$T = T_{on} + T_{off}$ = chopping period.

$$\alpha = \frac{T_{on}}{T} = \text{duty cycle.}$$

- Thus load voltage can be controlled by varying duty cycle α .

$$V_o = f \cdot T_{on} \cdot V_s$$

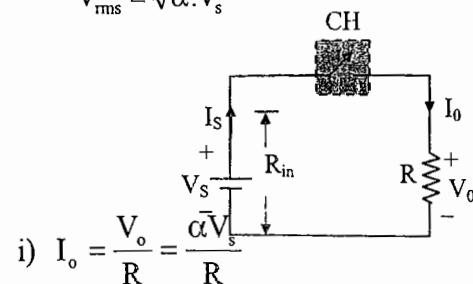
Where, $f = \frac{1}{T}$ = chopping frequency.

- Variation of T_{on} means adjustment of pulse width, as such this scheme is also called pulse-width modulation scheme.

- Average output current, $I_o = \frac{V_o}{R} = \alpha \frac{V_s}{R}$

- Rms value of output voltage = $\left[\frac{T_{on}}{T} \cdot V_s^2 \right]^{1/2}$

$$V_{rms} = \sqrt{\alpha} \cdot V_s$$

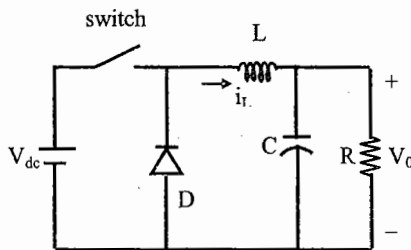


$$I_s(\text{avg}) = \frac{\alpha V_s}{R}$$

$$\text{ii) } R_{in} = \frac{V_s}{I_s} = \frac{V_s}{\frac{\alpha V_s}{R}} = \frac{R}{\alpha}$$

Worked out Examples

01. A buck converter circuit is shown in the figure. V_{dc} is input voltage and V_0 is the output voltage. Consider i_L and V_0 as state variables.



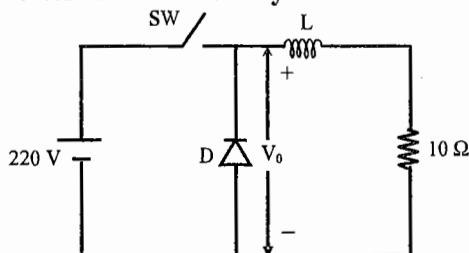
If $R = 1\Omega$, $L = 1H$ and $C = 0.1 F$ then the nature of the response will be

- Under damped
- Critically damped
- Over damped**
- Undamped

Sol: Eigen values of state matrix are -1.127 and -8.87

$\therefore$ Response is over damped

02. The following chopper circuit is operating at a switching frequency of 1 kHz with a duty cycle ratio of 50% . Assume a voltage drop of $2V$ across the switch when it is ON. Find the converter circuit efficiency.



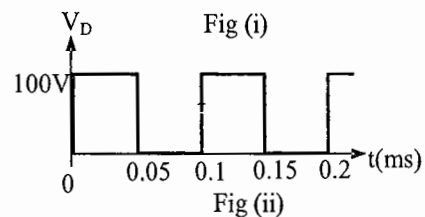
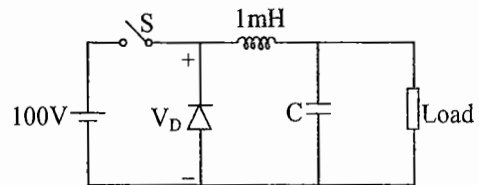
- 95%
- 98%
- 100%
- 99%**

$$\begin{aligned} \text{Sol: } P_0 &= \frac{1}{T} \int_0^T \frac{V_0^2}{R} dt = \frac{1}{T} \int_0^T \frac{(V_{dc} - V_{sw})^2}{R} dt \\ &= \frac{D(V_{dc} - V_{sw})^2}{R} \\ &= 2376.2 \text{ w} \end{aligned}$$

$$\begin{aligned} P_i &= \frac{1}{T} \int_0^T \frac{V_0^2}{R} dt = \frac{1}{T} \int_0^{DT} \frac{(V_{dc} - V_{sw})^2}{R} dt \\ &= 2398 \text{ w} \end{aligned}$$

$$\eta = 99\%$$

03. Fig. (i) Shows the circuit diagram of chopper. The switch is in the circuit in figure (i) is switched such that the voltage V_D across the diagram as the wave shape as show in figure (ii). The capacitance C is large so that the voltage across it is constant. If switch 'S' and diode are ideal, the peak to peak ripple (in A) in the inductor current is _____ (GATE-14)

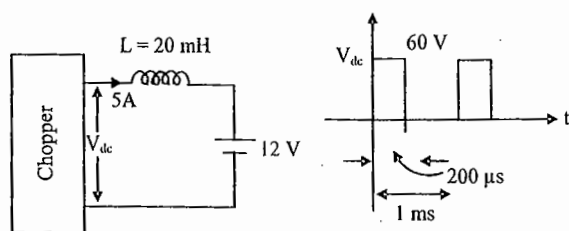


Sol:

The given circuit is buck converter peak to peak ripple in inductor current

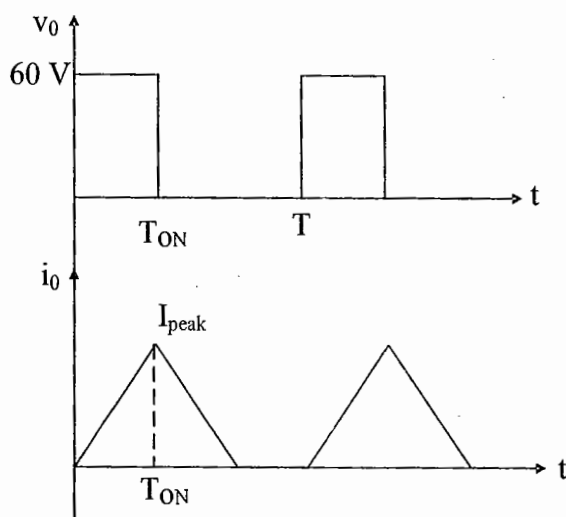
$$\begin{aligned} \Delta I_L &= \frac{V_{dc}}{L} D(1-D)T \\ \Rightarrow \Delta I_L &= \frac{100}{1 \times 10^{-3}} \times 0.5 \times 0.5 \times (0.1 \times 10^{-3}) \\ &= \frac{100 \times 0.1}{4} = 2.5A \end{aligned}$$

04. A chopper is employed to charge a battery as shown in figure. The charging current is $5 A$. The duty ratio is 0.2 . The chopper output voltage is also shown in figure. The peak to peak ripple current in the charging current is _____ (GATE-03)



- (a) 0.48 A (b) 1.2 A
(c) 2.4 A (d) 1 A

Sol: Given $\alpha = 0.2$, $T_{ON} = 200 \mu s$,
 $T = 1 ms$



A constant dc voltage applied to an inductor then nature of current is linear.

$$i = \left(\frac{v_s - E}{L} \right) t$$

$$I_p = \left(\frac{v_s - E}{L} \right) T_{ON}$$

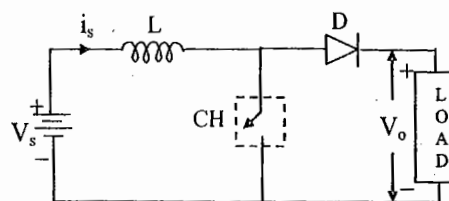
$$= \left(\frac{60 - 12}{20 \times 10^{-3}} \right) \times 0.2 \times 10^{-3} = 0.48 A$$

b) What type of chopper it is?

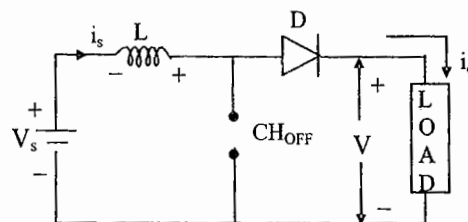
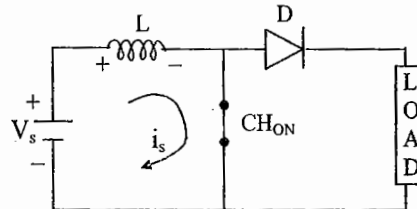
Ans: Step down Chopper.

Principle of operation of STEP – UP chopper:

- When the average output voltage V_o greater than input voltage V_s , then the chopper is called step up chopper.



- In this chopper, a large inductor L in series with source voltage V_s is essential as shown in fig. when the chopper CH is on, the closed current path is as shown in fig.



and inductor stores energy during T_{on} period. When the chopper CH is off, as the inductor current cannot die down instantaneously this current is forced to flow through the diode and load for a time T_{off} . As the current tends to decrease, polarity of the emf induced in L is reversed as shown.

- If dc is given to pure inductor, then nature of current is linear and slope of variation is $\left(\frac{V_s}{L} \right)$.
- Initially it is open circuited, finally it is short circuited.
- As a result, voltage across the load, given by $V_o = V_s + L \left(\frac{di}{dt} \right)$, exceeds the source voltage V_s .
- When CH is on, current through inductor ' L ' increase from I_1 to I_2 and when CH is off, current would fall from I_2 to I_1 .
- When CH is on, source voltage is applied to L i.e $V_L = V_s$.
When CH is off, KVL gives

$$V_L - V_o + V_s = (\text{or}) V_L = (V_o - V_s).$$

- The energy input to inductor from the source, during the period T_{on} , is
- $W_{in} = (\text{Voltage across } L) (\text{average current through } L) T_{on}$

$$W_{in} = V_s \left(\frac{I_1 + I_2}{2} \right) T_{on}.$$

- During the time T_{off} , when chopper is off, the energy released by inductor to the load is
- $W_{off} = (\text{Voltage across } L) (\text{average current through } L) T_{off}$

$$= (V_o - V_s) \left(\frac{I_1 + I_2}{2} \right) T_{off}$$

- Considering the system to be lossless, these two energies will be equal.

$$\therefore V_s \left(\frac{I_1 + I_2}{2} \right) T_{on} = (V_o - V_s) \left(\frac{I_1 + I_2}{2} \right) T_{off}$$

$$V_s \cdot T_{on} = (V_o - V_s) T_{off}$$

$$V_o T_{off} = V_s (T_{on} + T_{off}) = V_s \cdot T$$

$$V_o = V_s \frac{T}{T_{off}} = V_s \frac{T}{T - T_{on}} = V_s \frac{1}{1 - \alpha}$$

$$V_o = \left(\frac{1}{1 - \alpha} \right) V_s$$

Workedout Examples

05. A step-up chopper has an input voltage of 200 V and an output voltage of 500V. The ON time of the chopper is 105 μsec . Time period of the chopper (in μsec) is

- (a) 70 (b) 105 (c) 175 (d) 280

Sol: For a step up chopper, it can be shown that V_o = average load voltage

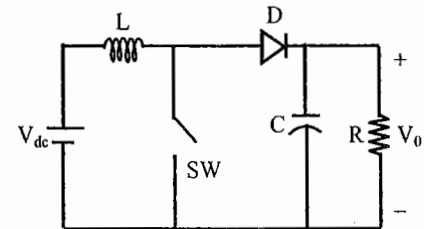
$$= \text{source voltage} \left(\frac{1}{1 - \alpha} \right)$$

Here, $500 = 200 \left(\frac{1}{1 - \alpha} \right)$; from which α the

duty ratio = 0.6. Since $T_{ON} = 105 \mu\text{s}$;

$$T = 175 \mu\text{s}$$

06. A boost converter is shown in the figure. The DC source voltage is 100 V and load resistance is 10Ω . Assume that the inductor has an internal resistance of 0.5Ω . The range of duty cycle which can give the stable operation of the converter circuit is

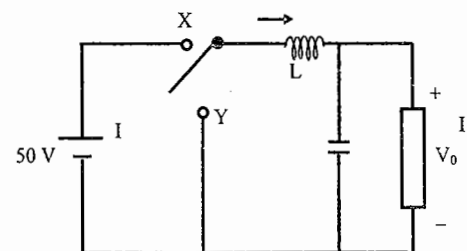


- (a) $0 < D < 0.78$ (b) $0.78 < D < 1$
(c) $0.8 < D < 1$ (d) $0 < D < 1$

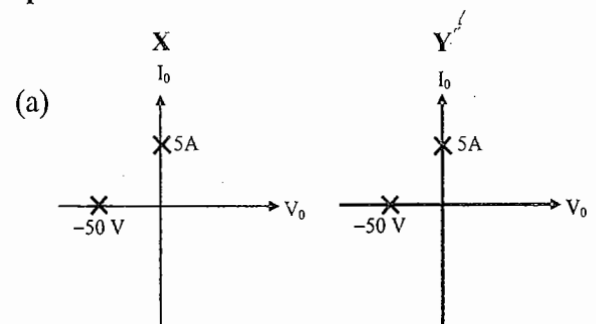
Sol: Boost converter is stable when $0 < D < D_{max}$

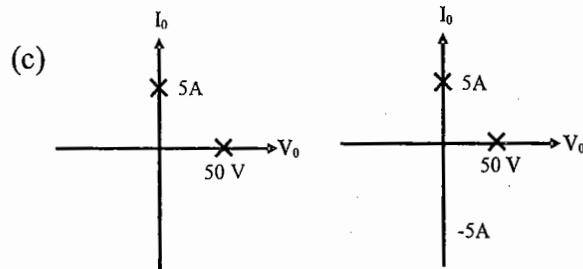
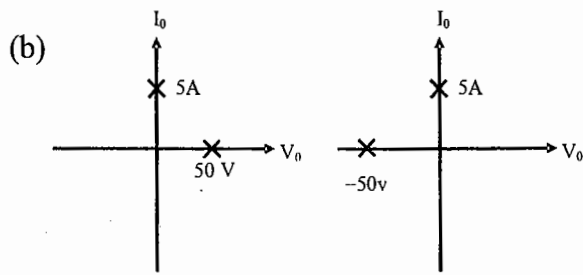
$$D_{max} = 1 - \sqrt{\frac{r}{R}} = 0.78$$

07. A Power converter is shown in the figure has two power switching devices namely X and Y. The source voltage is 50 V. The inductor current is steady 5A without any ripple.



On the V-I plane, identify the correct operating points of switches from the given options.

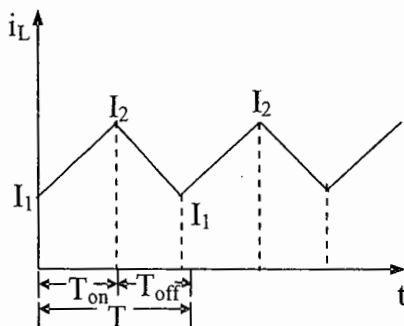
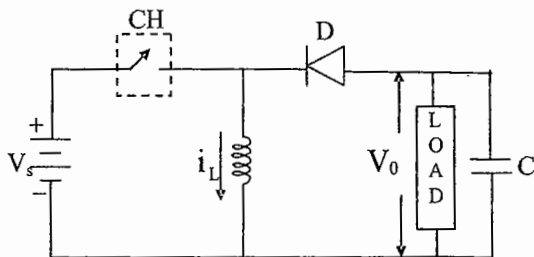




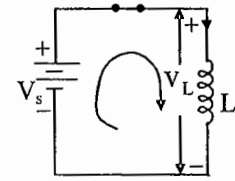
(d) None

Sol: X corresponds to BJT
Y corresponds to Diode

STEP DOWN/STEP UP CHOPPERS:



- When chopper is ON, current flows from source V_s to CH, L and back to V.
- Energy stored in inductor during T_{on} , $W_{in} = V_s \left(\frac{I_1 + I_2}{2} \right) T_{on}$

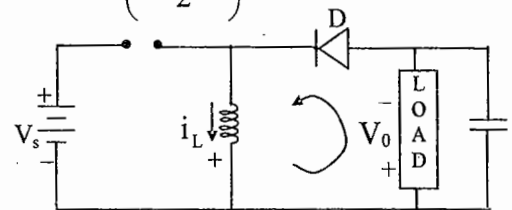


- When chopper is OFF, inductor current tends to decrease.

by KVL $V_L - V_o = 0$

$$V_L = V_o.$$

$$W_{OFF} = V_o \left(\frac{I_1 + I_2}{2} \right) T_{OFF}$$



- Inductor stored energy now discharges through the path; Load, diode D and L during the time chopper is OFF.
- Assuming the system to be lossless, the energy balance equation

$$W_{in} = W_{off}$$

$$V_s T_{ON} = V_o T_{OFF}$$

$$V_o = V_s \frac{T_{ON}}{T_{OFF}}$$

$$V_o = V_s \left(\frac{\alpha}{1 - \alpha} \right)$$

for step down chopper, $V_o = \alpha V_s$

for step up chopper, $V_o = \left(\frac{1}{1 - \alpha} \right) V_s$

for step up/ step down chopper,

$$V_o = \left(\frac{\alpha}{1 - \alpha} \right) V_s$$

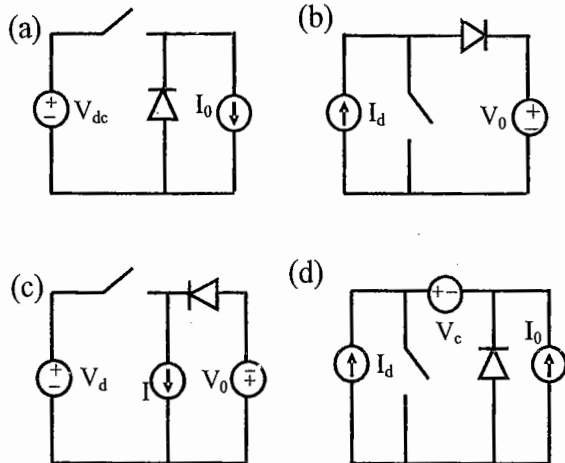
for $0 < \alpha < 0.5$, circuit works as step down chopper.

for $0.5 < \alpha < 1$, it operates as a step - up chopper.



Workedout Examples

08. Which of the following converter circuit representation is equivalent to Buck-Boost converter.



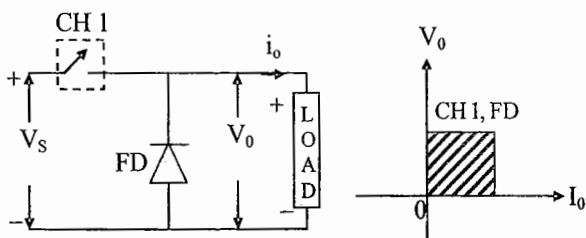
Ans: (c)

09. Regarding Buck-Boost converter, which of the statement is true

1. It will operate as Buck converter when $0 < D < 0.5$
2. It will operate as Boost converter when $0.5 < D < 1.0$ and stable throughout this range
3. It will operate as Boost converter when $0.5 < D < D_{max}$ and unstable if $D > D_{max}$.

- (a) only 1 & 3 are true
- (b) all are true
- (c) only 1&2 are true
- (d) None of these

First-Quadrant or Type- A Chopper:



- When CH 1 is on, $V_o = V_s$ and current i_o flows in positive direction.
- When CH 1 is off, $V_o = 0$ but i_o in the load continues flowing in the same direction through free wheeling diode FD.

- It is seen that average values of V_o and I_o are always positive.
- This chopper is suitable for only Motoring Applications.
- This chopper is also called step down chopper.

Workedout Example

10. A battery with its terminal voltage of 200 V is supplied power with type – A chopper circuit. The output voltage of the chopper consists of rectangular pulses of 2 ms duration in an overall cycle time of 5 ms. Internal resistance of the battery is negligible. Calculate:

- (a) Ripple factor
- (b) average and rms values of output voltage
- (c) rms value of the fundamental component of the output voltage
- (d) ac ripple voltage

Sol: Type – A, $V_s = 200V$, $T_{ON} = 2ms$ and $T = 5 ms$

$$\therefore \alpha = \frac{2}{5} = 0.4$$

$$(a) R.f = \sqrt{\frac{1}{\alpha} - 1} = \sqrt{\frac{1}{0.4} - 1} = 1.224$$

$$(b) V_o = \alpha V_s = 0.4 \times 200 = 80 V$$

$$V_{or} = \sqrt{\alpha} V_s = \sqrt{0.4} \times 220 \Rightarrow 126.5V$$

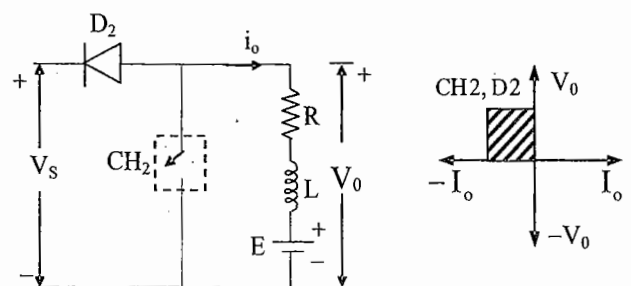
$$(c) V_{o1} = \left\{ \frac{2V_s}{\pi} \sin(\pi\alpha) \right\} \frac{1}{\sqrt{2}}$$

$$= 85.63 V$$

$$(d) R.f = \frac{V_{ac}}{V_{dc}}$$

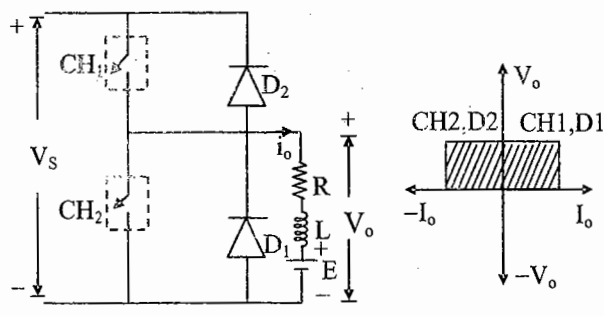
$$V_{ac} = 1.224 \times 80 = 97.98V$$

Second – Quadrant (or) Type – B Chopper:



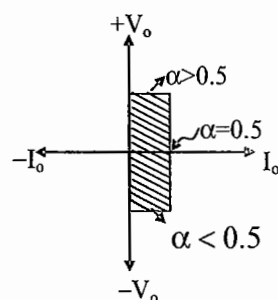
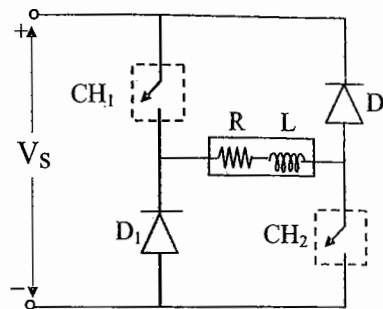
- In this chopper load must be of RLE in nature.
- When CH2 is on, $V_o = 0$ but load voltage E drives current through L and CH2.
- Inductor L stores energy during T_{on} of CH2.
- When CH2 is off, $V_o = \left[E + L \frac{di}{dt} \right]$ exceeds source voltage V_s .
- As a result, diode D_2 is forward biased and begins conduction.
- Chopper CH2 may be on or off, current I_o flows out of the load, current i_o is there fore treated as negative.
- The power flow is always from load to source. Type B chopper is also called step up chopper.
- In Regenerative braking Applications, step up choppers formula is to be employed.

Type C Chopper (or) Two Quadrant Type A Chopper:



- This type of chopper is obtained by connecting type A and type B choppers in parallel as shown in fig.
- The output voltage V_o is always positive because of the presence of freewheeling diode D_1 across the load.
- When CH2 is on, or Freewheeling diode D_1 conducts, output voltage $V_o = V_s$.
- The load current i_o can, however, reverse its direction. load current is positive when CH1 is on or FD conducts. Load current is negative if CH2 is on or D_2 conducts.
- Average load voltage is always positive but average load current may be positive or negative.

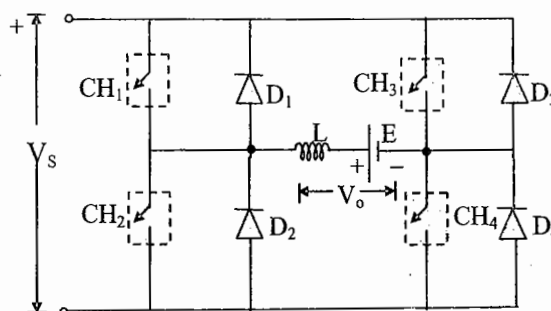
Type D Chopper, (or) Two Quadrant Type B Chopper:



- The output voltage $V_o = V_s$ when both CH1 and CH2 are ON. and $V_o = -V_s$ when both choppers are off but both diodes D_1 and D_2 conduct.
- The direction of Load current is always positive.
- In case $T_{on} > T_{off}$, $\alpha > 0.5$, V_o is positive.
- In case $T_{on} < T_{off}$, $\alpha < 0.5$, V_o is negative.
- In case $T_{on} = T_{off}$, $\alpha = 0.5$, V_o is zero.

Four Quadrant Chopper or Type E Chopper: First Quadrant:

- For this operation CH_4 is kept on, CH_3 is kept off and CH_1 is operated.
- With CH_1 , CH_4 on, load voltage $V_o = V_s$ and load current i_o begins to flow.
- When CH_1 is turned off, positive current free wheels through CH_4 , D_2 .

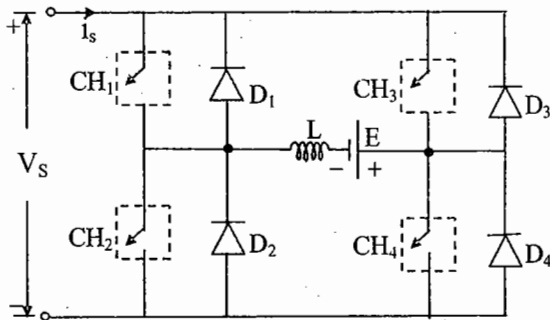


Second Quadrant:

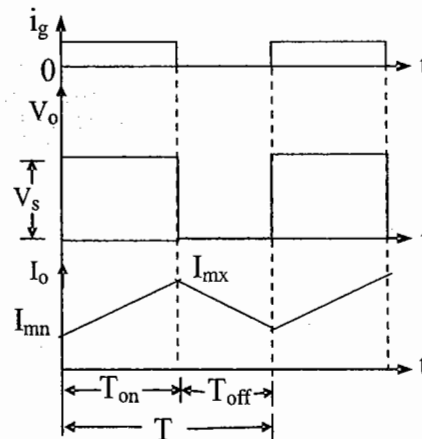
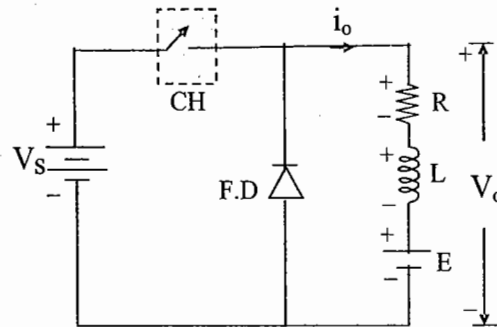
- Here CH2 is operated and CH1, CH3, and CH4 are kept off.
- with CH2 on, reverse current (negative) flows through L, CH2, D4 and E. Inductor L stores energy during CH2 is on. when CH2 is turned off, current is fed back to source through diodes D1, D4. Note that here $\left(E + L \frac{di}{dt}\right)$ is more than the source voltage V_s .

Third Quadrant:

- For this operation, CH1 is kept off, CH2 is kept on and CH3 is operated. Polarity of load emf E must be reversed for this quadrant working.
- CH3 is on, load gets connected to source V_s so that both V_o , i_o are negative leading to third quadrant operation.
- When CH3 is off, negative current freewheels through CH2, D4.


Fourth Quadrant:

- Here CH4 is operation and other are kept OFF. Load emf E has its polarity as in 3rd quadrant. With CH4 ON, positive current flows through CH4, D2, L and E. inductor L stores energy during the time CH4 is on.
- When CH4 is turned OFF, current is Fed back to source through diodes D2,D3. Here load voltage is negative, but load current is positive. All power is fed back from load to source.

Steady State Analysis of Type – A Chopper:


- Inductor is assumed to be having the sufficient energy to support the conduction through out half time interval,

$$V_o = \alpha V_s \text{ (Average output voltage)}$$

$$V_{or} = \sqrt{\alpha} V_s$$

(Rms value of output voltage)

$$I_{mx} = \frac{V_s}{R} \left[\frac{1 - e^{-T_{on}/T_a}}{1 - e^{-T/T_a}} \right] - \frac{E}{R}$$

$$I_{mn} = \frac{V_s}{R} \left[\frac{e^{T_{on}/T_a} - 1}{e^{T/T_a} - 1} \right] - \frac{E}{R}$$

$$\text{Where, } T_a = \text{Load time constant} = \frac{L}{R}$$

$$\text{Current Ripple, } (\Delta I) = I_{mx} - I_{mn}$$

$$\Delta I = \frac{V_s}{R} \left[\left(\frac{1 - e^{-T_{on}/T_a}}{1 - e^{-T/T_a}} \right) - \left(\frac{e^{T_{on}/T_a} - 1}{e^{T/T_a} - 1} \right) \right]$$

$$= \frac{V_s}{R} \left[\frac{1 - e^{-T_{on}/T_a}}{1 - e^{-T/T_a}} - \frac{e^{T_{on}/T_a} (1 - e^{-T_{on}/T_a})}{e^{T/T_a} (1 - e^{-T/T_a})} \right]$$

$$= \frac{V_s}{R} \left[\frac{1 - e^{-T_{on}/T_a}}{1 - e^{-T/T_a}} - \frac{e^{-(T-T_{on})/T_a} (1 - e^{-T_{on}/T_a})}{(1 - e^{-T/T_a})} \right]$$

$$= \frac{V_s}{R} \left[\frac{(1 - e^{-T_{on}/T_a}) - e^{-T_{off}/T_a} (1 - e^{-T_{on}/T_a})}{(1 - e^{-T/T_a})} \right]$$

$$\Delta I = \frac{V_s}{R} \left[\frac{(1 - e^{-T_{on}/T_a})(1 - e^{-T_{off}/T_a})}{1 - e^{-T/T_a}} \right]$$

$$\text{Per unit Ripple} = \frac{\Delta I}{V_s / R}$$

Per unit Ripple =

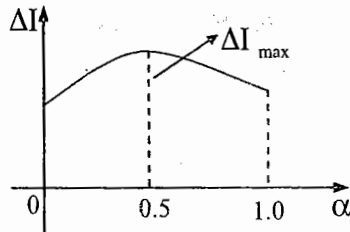
$$\frac{\Delta I}{V_s / R} = \left[\frac{(1 - e^{-T_{on}/T_a})(1 - e^{-T_{off}/T_a})}{(1 - e^{-T/T_a})} \right]$$

$$T_{on} = \alpha T$$

$$T_{OFF} = (1 - \alpha)T$$

$$\Delta I = \frac{V_s}{R} \left[\frac{(1 - e^{-\alpha T/T_a})(1 - e^{-(1-\alpha)T/T_a})}{(1 - e^{-T/T_a})} \right]$$

- Per unit Ripple (or) Ripple is a function of duty cycle ' α '.



- Ripple is minimum at $\alpha = 0$, increases maximum at $\alpha = 0.5$ and decreases at $\alpha = 1.0$.
- For $\alpha = 0.5$, Ripple would be maximum.

$$(\Delta I)_{max} = \frac{V_s}{R} \left(\frac{(1 - e^{-0.5x})(1 - e^{-0.5x})}{1 - e^{-x}} \right) \Bigg|_{\substack{\text{Let} \\ \frac{T}{T_a} = x}}$$

$$= \frac{V_s}{R} \left(\frac{(1 - e^{-0.5x})(1 - e^{-0.5x})}{(1 + e^{-0.5x})(1 - e^{-0.5x})} \right)$$

$$= \frac{V_s}{R} \tanh(0.25x)$$

$$= \frac{V_s}{R} \tanh\left(\frac{1}{4} \times \frac{T}{T_a}\right)$$

$$= \frac{V_s}{R} \tanh\left(\frac{1}{4} \times \frac{1}{f} \times \frac{R}{L}\right)$$

$$(\Delta I)_{max} = \frac{V_s}{R} \tanh\left(\frac{R}{4fL}\right)$$

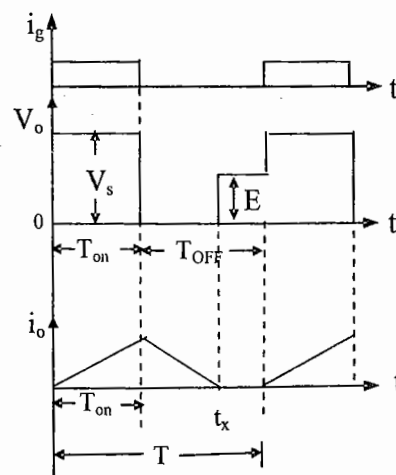
- In case $4fL \gg R$, then $\tanh\left(\frac{R}{4fL}\right) \cong \left(\frac{R}{4fL}\right)$ under this condition maximum value of ripple current is $\Delta I_{mx} = \frac{V_s}{R} \cdot \frac{R}{4fL} = \frac{V_s}{4fL}$

$$\Delta I_{mx} = \frac{V_s}{4fL}$$

- This shows that maximum value of ripple current is inversely proportional to chopping frequency and the circuit inductance.
- For the better performance of the load, the conduction to be continuous. It is preferable to have the less value of the current ripple (or) More uniform output current.
- The chopping frequency is to be high. It is possible by employing the smaller values of time period.

Discontinuous Conduction:

- If the inductor is not having the sufficient energy to support the conduction through out OFF time interval, then the load current becomes discontinuous. During this period current is zero, out put voltage is equal to E.
- The respective wave forms are as shown in fig.



- The average output voltage for the discontinuous current mode as shown in fig.



$$V_o = \frac{1}{T} \int_0^T V_o dt$$

$$= \frac{1}{T} \left[\int_0^{T_{on}} V_s dt + \int_{T_{on}}^{t_x} 0 dt + \int_{t_x}^T E dt \right]$$

$$= V_s \frac{T_{on}}{T} + E \left(\frac{T - t_x}{T} \right)$$

$$V_o = \alpha V_s + E \left(1 - \frac{t_x}{T} \right) \text{ (volts)}$$

- Average value of out put voltage will be more in Discontinuous conduction compared to continuous conduction.
- Fourier Analysis of output voltage (continuous conduction) :

$$V_o = V_0 + \sum_{n=1,2}^{\infty} V_n$$

Where, $V_0 = \alpha V_s$ (D.C component)

Rms value of fundamental component

$$V_{01} = \left\{ \frac{2V_s}{\pi} \sin(\pi\alpha) \right\} \frac{1}{\sqrt{2}}$$

AC ripple voltage V_r .

$$V_r = \sqrt{V_{rms}^2 - V_0^2}$$

$$V_{rms} = V_{or}$$

= is the rms value of output voltage.

V_0 = average value of output voltage.

$$V_{or} = \sqrt{\alpha} V_s$$

$$V_o = \alpha V_s$$

$$V_r = \sqrt{\alpha V_s^2 - \alpha^2 V_s^2}$$

$$V_r = V_s \sqrt{\alpha - \alpha^2}$$

Ripple factor (RF): It is defined as the ratio of ac ripple voltage to average voltage, is given

$$\text{by } RF = \frac{V_r}{V_o}$$

$$R.F. = \frac{V_s \sqrt{\alpha - \alpha^2}}{V_s \cdot \alpha} = \sqrt{\frac{1 - \alpha}{\alpha}} = \sqrt{\frac{1}{\alpha} - 1}$$

- Chopper operating principle is based on Forced Commutation. There are two groups of commutation.
 1. Forced commutation.
 2. Load Commutation.
- Forced Commutation is of two types:
 1. Voltage Commutation.
 2. Current Commutation.

Both the above commutation circuit employ tank circuit for their operation. Tank circuit consists of Inductor, capacitor and unidirectional device (diode (or) Thyristor).